

Data Communications & Networking

PROJECT 1

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The image shows a Wireshark packet capture window titled 'ip-wireshark-trace1-1.pcapng'. The packet list on the left shows several packets, with packet 44 selected. The packet details pane on the right shows the structure of the selected packet, which is an Internet Protocol Version 4 (IPv4) packet. The packet is 70 bytes long and is a standard query response from 192.168.86.1 to 128.119.245.12. The packet details pane shows the following information:

- Frame 44: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface 0
- Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:4f:43:98:d9:27
- Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 56
 - Identification: 0xfda1 (64929)
 - 000. = Flags: 0x0
 - ...0 0000 0000 0000 = Fragment Offset: 0

The packet bytes pane on the right shows the raw data of the packet in hexadecimal and ASCII format.

1. Select the first UDP segment sent by your computer via the traceroute command to gaia.cs.umass.edu. (Hint: this is 44th packet in the trace file in the ip_wireshark-trace1-1.pcapng file). Expand the Internet Protocol part of the packet in the packet details window. What is the IP address of your computer?

A. IP address - 192.168.86.61

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> Frame 44: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on
> Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:06:00:00:00:00
✓ Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    ✓ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport
    Total Length: 56
    Identification: 0xfda1 (64929)
    > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    ✓ Time to Live: 1
        > [Expert Info (Note/Sequence): "Time To Live" only 1]
        Protocol: UDP (17)
        Header Checksum: 0x2faa [validation disabled]
        [Header checksum status: Unverified]
        Source Address: 192.168.86.61

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2. What is the value in the time-to-live (TTL) field in this IPv4 datagram's header?

A. TTL – 1

3. What is the value in the upper layer protocol field in this IPv4 datagram's header? [Note: the answers for Linux/MacOS differ from Windows here].

A. UDP (17)

4. How many bytes are in the IP header?

A. 20 Bytes

5. How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes.

A. Total length – header length = 56 – 20 = 36 Bytes

6. Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented

A. The More Fragments bit is not set, indicating that this is the last fragment or that the packet is not fragmented at all.

7. Which fields in the IP datagram always change from one datagram to the next within this series of UDP segments sent by your computer destined to 128.119.245.12, via traceroute? Why? (List ALL the fields that change)

**A. TTL (time-to-live) - It is incremented by traceroute with each datagram to discover path
 Identification field - Unique for every datagram.
 Checksum - Computed every time**

8. Which fields in this sequence of IP datagrams (containing UDP segments) stay constant? Why? (List ALL the fields that stay constant).

A. **Source address – Packets are sent from same source (my machine).**

Destination address – Packets are sent to same destination.

Protocol – UDP here.

Header Length – each IP header is of same size.

Flags - since, these are not fragmented don't fragment and more will always 0.

9. Describe the pattern you see in the values in the Identification field of the IP datagrams being sent by your computer.

A. **identification fields are assigned incrementally (by 1) to each individual datagram.**

10. What is the upper layer protocol specified in the IP datagrams returned from the routers? [Note: the answers for Linux/MacOS differ from Windows here].

A. **ICMP**

11. Are the values in the Identification fields (across the sequence of all of ICMP packets from all of the routers) similar in behavior to your answer to question 9 above?

A. **Here, identification fields are random and not in similar behavior to question 9.**

12. Are the values of the TTL fields similar, across all of ICMP packets from all of the routers?

A. **TTL =1 for all.**

13. Find the first IP datagram containing the first part of the segment sent to 128.119.245.12 sent by your computer via the traceroute command to gaia.cs.umass.edu, after you specified that the traceroute packet length should be 3000. (Hint: This is packet 179 in the ip-wireshark-trace1-1.pcapng trace file.). Has that segment been fragmented across more than one IP datagram?

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
171	10.244346	128.119.240.65	192.168.86.61	TLSv1.2	102	Application Data
172	10.244350	128.119.240.65	192.168.86.61	TLSv1.2	1150	Application Data
173	10.244440	192.168.86.61	128.119.240.65	TCP	66	57262 → 443 [ACK] Seq=1 Ack=37 Win=2047 Len=0 TSv
174	10.244440	192.168.86.61	128.119.240.65	TCP	66	57262 → 443 [ACK] Seq=1 Ack=1121 Win=2030 Len=0 T
175	10.262582	52.114.132.176	192.168.86.61	TLSv1.2	388	Application Data
176	10.262654	192.168.86.61	52.114.132.176	TCP	54	56197 → 443 [ACK] Seq=1 Ack=335 Win=4090 Len=0
177	10.289567	192.168.86.61	52.114.132.176	TLSv1.2	242	Application Data
178	10.370823	52.114.132.176	192.168.86.61	TCP	60	443 → 56197 [ACK] Seq=335 Ack=189 Win=2053 Len=0
179	12.788154	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=0, ID=f...
180	12.788155	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=1480, I

Identification: 0xfda2 (64930)

- 001. = Flags: 0x1, More fragments
 - 0... = Reserved bit: Not set
 - 0... = Don't fragment: Not set
 - 1... = More fragments: Set
- ...0 0000 0000 0000 = Fragment Offset: 0
- > Time to Live: 1
- Protocol: UDP (17)
- Header Checksum: 0xa0a5 [validation disabled]
- [Header checksum status: Unverified]

0010 05 dc fd a2 26 00 01 11 0a 0
0020 f5 0c fd a1 82 9b 0b a4 db f
0030 00 00 00 00 00 00 00 00 0
0040 00 00 00 00 00 00 00 00 0
0050 00 00 00 00 00 00 00 00 0
0060 00 00 00 00 00 00 00 00 0
0070 00 00 00 00 00 00 00 00 0
0080 00 00 00 00 00 00 00 00 0
0090 00 00 00 00 00 00 00 00 0
00a0 00 00 00 00 00 00 00 00 0
00b0 00 00 00 00 00 00 00 00 0
00c0 00 00 00 00 00 00 00 00 0

A. **Yes, the segment has been fragmented across more than 1 IP datagram.**

14. What information in the IP header indicates that this datagram been fragmented?

A. **More Fragment is set to 1 which indicates that this datagram has been fragmented.**

- A. Fragment offset here is 0, which means this is the first fragment.

- A.

[illegible]

- A. $1480(\text{Data}) + 20 (\text{IP header}) = 1500 \text{ Bytes}$**

- What information in the IP header indicates that this is not the first datagram fragment?

No.	Time	Source	Destination	Protocol	Length	Info
176	10.262654	192.168.86.61	52.114.132.176	TCP	54	56197 → 443 [ACK] Seq=1 Ack=335 Win=4090 Len=0
177	10.289567	192.168.86.61	52.114.132.176	TLSv1.2	242	Application Data
178	10.370823	52.114.132.176	192.168.86.61	TCP	60	443 → 56197 [ACK] Seq=335 Ack=189 Win=2053 Len=0
179	12.788154	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=0, ID=f)
180	12.788155	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=1480, I
181	12.788155	192.168.86.61	128.119.245.12	UDP	54	64929 → 33435 Len=2972
182	12.792190	192.168.86.1	192.168.86.61	ICMP	590	Time-to-live exceeded (Time to live exceeded in t
183	12.792881	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=0, ID=f
184	12.792882	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=1480, I
185	12.792882	192.168.86.61	128.119.245.12	UDP	54	64929 → 33436 Len=2972

001. = Flags: 0x1, More fragments
 0... = Reserved bit: Not set
 .0.. = Don't fragment: Not set
 ..1. = More fragments: Set
 ...0 0000 1011 1001 = Fragment Offset: 1480
 > Time to Live: 1
 Protocol: UDP (17)
 Header Checksum: 0x094c [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 192.168.86.61

A. Fragment offset here is 1480 which means this is not first fragment(as the first fragment offset will be 0)

18. What fields change in the IP header between the first and second fragment? List ALL the fields that change in the IP header.

A. Fragment offset and Header Checksum.

19. Now find the IP datagram containing the third fragment of the original UDP segment. What information in the IP header indicates that this is the last fragment of that segment?

No.	Time	Source	Destination	Protocol	Length	Info
176	10.262654	192.168.86.61	52.114.132.176	TCP	54	56197 → 443 [ACK] Seq=1 Ack=335 Win=4090 Len=0
177	10.289567	192.168.86.61	52.114.132.176	TLSv1.2	242	Application Data
178	10.370823	52.114.132.176	192.168.86.61	TCP	60	443 → 56197 [ACK] Seq=335 Ack=189 Win=2053 Len=0
179	12.788154	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=0, ID=f)
180	12.788155	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=1480, I
181	12.788155	192.168.86.61	128.119.245.12	UDP	54	64929 → 33435 Len=2972
182	12.792190	192.168.86.1	192.168.86.61	ICMP	590	Time-to-live exceeded (Time to live exceeded in t
183	12.792881	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=0, ID=f
184	12.792882	192.168.86.61	128.119.245.12	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=1480, I
185	12.792882	192.168.86.61	128.119.245.12	UDP	54	64929 → 33436 Len=2972

000. = Flags: 0x0
 0... = Reserved bit: Not set
 .0.. = Don't fragment: Not set
 ..0. = More fragments: Not set
 ...0 0001 0111 0010 = Fragment Offset: 2960
 > Time to Live: 1
 Protocol: UDP (17)
 Header Checksum: 0x2e47 [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 192.168.86.61

A. More Fragment here is 0 which means there are no more fragments and is a last fragment.